Lecture Note 8: Divisibility and Primes

Foundations of Mathematics

1 Divisibility

Definition 1.1. Let a and b be integers with $a \neq 0$. We say that a *divides* b, and write $a \mid b$, if there is an integer k such that $b = ak$.

Example 1.2. We have $3 \mid 12$ because $12 = 3 \cdot 4$, but $5 \nmid 12$ because no integer k satisfies $12 = 5k$.

Lemma 1.1. *If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.*

Proof. Write $b = ak$ and $c = al$ with integers k and l. Then $b + c = a(k + l)$, and $k + l$ is an integer. □

Theorem 1.2 (Transitivity)**.** *If $a \mid b$ and $b \mid c$, then $a \mid c$.*

Proof. By definition there are integers k and l with $b = ak$ and $c = bl$. Substituting the first equation into the second gives

$$c = (ak)l = a(kl).$$

Since kl is an integer, $a \mid c$. □

Corollary 1.2.1. *If $a \mid b$, then $a \mid bc$ for every integer c.*

Proof. Since $b \mid bc$, this follows from Theorem 1.2. □

2 Prime Numbers

Definition 2.1. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p.

Example 2.2. The first primes are $2, 3, 5, 7, 11$ and 13. The number 1 is not prime, and 2 is the only even prime.

Theorem 2.1 (Euclid)**.** *There are infinitely many prime numbers.*

Proof. Suppose that there are only finitely many primes $p_1, p_2, \ldots, p_n$, and consider

$$N = p_1 p_2 \cdots p_n + 1.$$

Some prime p_i divides N, and p_i also divides $p_1 p_2 \cdots p_n$. By Lemma 1.1 applied to N and $-p_1 p_2 \cdots p_n$, the prime p_i divides their sum, which is

$$N - p_1 p_2 \cdots p_n = 1.$$

This is impossible, because no prime divides 1. □

Remark. The proof of Theorem 2.1 does not say that N itself is prime. For example, $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \cdot 509$.